Adelic Line Bundles

From Canonical Heights to Arithmetic Positivity

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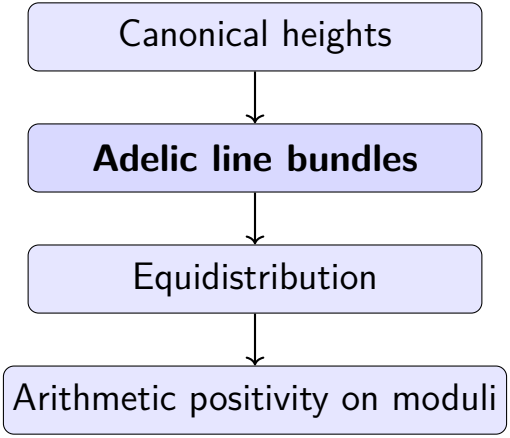
Arakelov and Diophantus

A guiding question

How should one understand canonical heights?

- **Before 1995:** Heights of points and algebraic cycles.
- **1995:** Canonical heights become degrees of adelic line bundles.
- **1997–2008:** Small points become equidistributed.
- **2021:** Canonical adelic geometry extends to quasi-projective varieties.
- **Today:** Arithmetic positivity on moduli.

Guiding theme



Height theory before adelic line bundles

Weil heights

- naive heights
- Weil heights
- Mordell–Weil theorem
- Roth theorem

Canonical heights

- Néron–Tate heights
- abelian varieties
- BSD conjecture

Arakelov heights

- hermitian line bundles
- arithmetic intersections
- Faltings' theorem

Common feature

Height theory measured the arithmetic complexity of **big points**.

The next step was to understand *small points*.

Weil heights

A height measures the arithmetic complexity of an algebraic point.

For a rational number

$$x = \frac{a}{b}, \qquad (a, b) = 1,$$

define

$$h(x) = \log \max\{|a|, |b|\}.$$

More generally, for

$$P = [x_0 : \cdots : x_n] \in \mathbf{P}^n(K),$$

over a number field K, the Weil height is obtained by combining the contributions from all places of K.

Arakelov geometry

Let $\mathcal{X}$ be an arithmetic variety over $\mathcal{O}_K$.
A hermitian line bundle

$$\overline{\mathcal{L}} = (\mathcal{L}, \|\cdot\|_\infty)$$

consists of

- a line bundle $\mathcal{L}$ on $\mathcal{X}$,
- hermitian metrics at the Archimedean places.

For a rational point x,

$$h_{\overline{\mathcal{L}}}(x) = \frac{1}{[K:\mathbf{Q}]}\widehat{\deg}\big(\overline{\mathcal{L}}|_x\big).$$

Arakelov principle

Classical heights are degrees of hermitian line bundles.

Adeles and product formula

Let K be a number field.
For every place v, let K_v be the completion of K.
The adele ring

$$\mathbf{A}_K = \prod_v{}' K_v$$

assembles all completions into a single global object.

Normalize each absolute value $|\cdot|_v$ so that

$$\mathrm{vol}(aU) = |a|_v \, \mathrm{vol}(U).$$

Then every nonzero element $a \in K^\times$ satisfies the

$$\boxed{\prod_v |a|_v = 1.}$$

Fundamental identity

The product formula is the fundamental identity relating all places of K.

Adelic line bundles

Let X/K be a projective variety.
An adelic line bundle is

$$\overline{L} = (L, \|\cdot\|_{\mathbf{A}_K}),$$

where L is a line bundle on X and $\|\cdot\|_{\mathbf{A}_K}$ is an adelic metric on $L_{\mathbf{A}_K}$.
Equivalently,

$$\|\cdot\|_{\mathbf{A}_K} \quad \Longleftrightarrow \quad \{\|\cdot\|_v\}_v$$

with a coherence condition: outside finitely many finite places, the metric is induced by a common integral model.

Adelic principle

Canonical heights are degrees of adelic line bundles.

Why “adelic”?

Class field theory

$$K \rightsquigarrow (K_v)_v \rightsquigarrow \mathbf{A}_K$$

Adeles assemble local fields into a single global object.

Adelic geometry

$$L \rightsquigarrow (\|\cdot\|_v)_v \rightsquigarrow (L, \|\cdot\|_{\mathbf{A}_K})$$

Adelic line bundles assemble local analytic metrics into a single global object.

Adelic line bundles not only enlarge hermitian line bundles; they bring analysis to every place.

Degree of an adelic line bundle at a point

Let $x \in X(K)$. The fiber L_x is a one-dimensional K-vector space with adelic metric.
For any nonzero $s \in L_x$, define

$$\boxed{\deg(\overline{L}_x) = -\log \|s\|_{\mathbf{A}_K}.}$$

This is independent of s by the product formula.
The associated height is

$$\boxed{h_{\overline{L}}(x) = \frac{1}{[K : \mathbf{Q}]} \deg(\overline{L}_x).}$$

Summary

Weil, Arakelov, and invariant heights all become degrees of metrized line bundles.

Weil height as an adelic degree

For $P = [x_0 : \cdots : x_n]$ and $s = \sum a_i X_i \in H^0(\mathbf{P}^n, \mathcal{O}(1))$,

$$\|s(P)\|_{\mathbf{A}_K} = \prod_v \frac{|\sum_i a_i x_i|_v}{\max_i |x_i|_v}.$$

By the product formula,

$$\deg\big(\overline{\mathcal{O}}(1)_p\big) = -\log \|s(P)\|_{\mathbf{A}_K} = \sum_v \log \max_i |x_i|_v.$$

$$\boxed{h(P) = \frac{1}{[K : \mathbf{Q}]} \deg\big(\overline{\mathcal{O}}(1)_P\big).}$$

Example: arithmetic dynamics

A polarized dynamical system is

$$(X, f, L), \qquad f^*L \simeq L^{\otimes q}, \quad q > 1.$$

It has

- a canonical semipositive adelic line bundle $\overline{L}_f$,
- the Call–Silverman height $\widehat{h}_f = h_{\overline{L}_f}$,

Three local languages

Complex geometry	**p-adic geometry**	**Adelic geometry**
Smooth metric	Integral model over $\mathbf{Z}_p$	Model/smooth adelic metric
Continuous psh metric	Nef integral model	Semipositive adelic metric
Difference of psh metrics	Difference of nef models	Integrable adelic metric
Monge–Ampère measure	Intersections with components over $\mathbf{F}_p$	Adelic Monge–Ampère measure

Philosophy

Adelic geometry is the global theory whose local manifestations are complex geometry and p-adic geometry.

Regularity conditions for adelic metrics

Semipositive

A semipositive adelic metric is the adelic analog of a metric with semipositive curvature:

- at Archimedean places, it is a continuous psh metric;
- at non-Archimedean places, it is a limit of metrics arising from nef models.

Integrable

An adelic line bundle is integrable if it is a difference of two semipositive adelic line bundles.

These are the natural classes for adelic intersection theory.

Adelic Monge–Ampère measures

Let $\overline{L}$ be a semipositive adelic line bundle on an n-dimensional projective variety X/K.

At every place v, a semipositive adelic metric determines a canonical Monge–Ampère measure

$$c_1(\overline{L})_v^n$$

on the analytic space X_v^{an}.

- At Archimedean places, this is the classical complex Monge–Ampère measure.
- At non-Archimedean places, it is constructed from arithmetic intersection theory and extended by continuity.

Adelic principle

The Monge–Ampère measure is defined uniformly at every place.

Equidistribution of small points

Let $\overline{L}$ be nef, with L big. Let (x_m) be a generic sequence of small points:

$$h_{\overline{L}}(x_m) \longrightarrow 0.$$

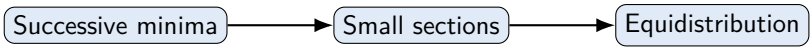
Then for every place v, the Galois orbits of x_m equidistribute:

$$\delta_{x_m,v} \longrightarrow \mu_{\overline{L},v} = \frac{c_1(\overline{L})_v^n}{\deg_L(X)}.$$

Small points become uniformly distributed at every place.

From Successive Minima to Equidistribution

- **1995 (Zhang):** Successive minima and arithmetic intersection theory of adelic line bundles.
- **1997 (Szpiro–Ullmo–Zhang):** Equidistribution at Archimedean places, assuming $c_1(\overline{L})_v$ is strictly positive.
- **2008 (Yuan):** Equidistribution at any place, without strict positivity, via the theory of big adelic line bundles.



The Bogomolov conjecture

Conjecture (Bogomolov)

Let X be a subvariety of an abelian variety.
If X is not a translate of an abelian subvariety by a torsion point, then the set of points of sufficiently small height is not Zariski dense.

Theorem

- Ullmo (1998): curves.
- Zhang (1998): arbitrary subvarieties.

Equidistribution provides the bridge from geometry to analysis.

Beyond projective varieties

Can adelic geometry be extended to quasi-projective varieties?

The projective theory does not immediately apply to

- moduli spaces,
- Shimura varieties,
- varieties over finitely generated fields.

These spaces play a central role in modern arithmetic geometry, but they naturally carry logarithmic singularities.

The next challenge

Develop a theory of adelic line bundles for quasi-projective varieties.

Adelic line bundles on quasi-projective varieties

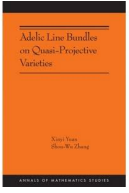
Joint work with Xinyi Yuan (2021)

A theory of adelic line bundles was developed for quasi-projective varieties by introducing

boundary topology.

The theory extends

- arithmetic intersection theory,
- positivity,
- equidistribution.



Projective varieties $\Longrightarrow$ quasi-projective varieties

Arithmetic moduli of curves

Let $\mathcal{M}_g$ be the moduli stack of smooth curves of genus g, and let

$$\pi : \mathcal{C} \longrightarrow \mathcal{M}_g \subset \overline{\mathcal{M}_g}$$

be the universal curve and ω the relative dualizing sheaf.
Important tautological line bundles on $\overline{\mathcal{M}_g}$:

$$\lambda = \det \pi_* \omega, \qquad \mu = \langle \omega, \omega \rangle.$$

Motivation

To study the arithmetic on $\mathcal{M}_g$ ($g \geq 2$), we need to understand two canonical adelic metrics on μ over $\mathcal{M}_g$.

Adelic Poincaré bundles

For simplicity, we replace $\mathcal{M}_g$ by the moduli $\mathcal{M}$, parametrizing pairs (C, ξ) with

$$(2g-2)\xi = \omega_C,$$

Let $\mathcal{J} \to \mathcal{M}$ be the universal Jacobian and

$$i : \mathcal{C} \longrightarrow \mathcal{J}, \qquad x \mapsto [x - \xi]$$

the Abel–Jacobi embedding.
Using the invariant adelic metric on the Poincaré bundle P on $\mathcal{J} \times \mathcal{J}$, one obtains adelic bundles

$$P_{\mathcal{C}\times\mathcal{C}} = (i \times i)^* P \qquad \text{on} \quad \mathcal{C} \times \mathcal{C},$$

$$P_{\mathcal{C}} = \Delta^* P_{\mathcal{C}\times\mathcal{C}} \qquad \text{on} \quad \mathcal{C}.$$

Double and triple products

Using Deligne's pairings, we obtain two adelic structures on μ:

$$\mu_a := \left(1 - \frac{1}{g}\right) \langle P_{\mathcal{C}}, P_{\mathcal{C}} \rangle,$$

$$\mu_b := \frac{2g-2}{2g+1} \langle P_{\mathcal{C}\times\mathcal{C}}, P_{\mathcal{C}\times\mathcal{C}}, P_{\mathcal{C}\times\mathcal{C}} \rangle.$$

For a curve $[C] \in \mathcal{M}_g(\bar{\mathbf{Q}})$:

- $h_{\mu_a}(C)$ measures the admissible height of C itself, and
- $h_{\mu_b}(C)$ measures the heights of the Gross–Schoen cycles or Ceresa cycles associated to C.

Yuan's positivity theorem

Theorem (Xinyi Yuan)

The adelic line bundle μ_a is nef and big.

One consequence is a uniform Bogomolov-type theorem: There exist constants $c_1, c_2 > 0$ depending only on g, such that for every curve $C/\overline{\mathbf{Q}}$ of genus g and every $\alpha \in \mathrm{Pic}^1(C)$,

$$\#\left\{x \in C(\overline{\mathbf{Q}}) : h_{NT}(x-\alpha) \leq c_1\big(h_{Fal}(C) + h_{NT}((2g-2)\alpha - \omega_C)\big)\right\} < c_2.$$

Gao–Zhang's positivity

When $g = 2$, μ_b is trivial.

Theorem (Gao–Zhang)

For $g \geq 3$, the generic fiber of μ_b is big.

Consequently, Gross–Schoen and Ceresa cycles satisfy a Northcott property on a nonempty open subset of $\mathcal{M}_g$: For $g \geq 3$, there is a non-empty open subset U of $\mathcal{M}_g$ such that for any positive numbers D, H,

$$\#\left\{x \in \mathcal{M}_g(\overline{\mathbf{Q}}) : \deg x \leq D, h_{\mu_b}(x) < H\right\} < \infty.$$

The evolution of adelic geometry

